

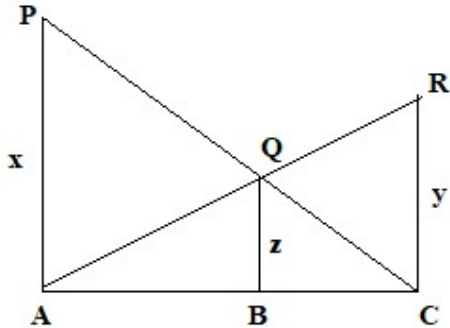
Pre – Board Examination (2024-25)

Class- X

Subject- Maths(Standard)– Marking Scheme

	Section A consists of 20 questions of 1 mark each.	
S.NO		Marks
1	(B) 13	1
2	(A) 18	1
3	(B) 15	1
4	(C) rational and distinct	1
5	(A) $\frac{11}{13}$	1
6	(B) 2	1
7	(A) $x^2 - 6x + 7$	1
8	(C) $45^0, 15^0$	1
9	(A) 3 cm	1
10	(B) -12	1
11	(D) 2	1
12	(A) (i) and (ii)	1
13	(B) 5	1
14	(D) can not be answered without knowing the value of R	1
15	(A) 179.66 cm^3	1
16	(B) 25	1
17	(A) 27	1
18	(C) $\frac{1}{25}$	1
19	(C) Assertion (A) is true but reason (R) is false.	1
20	(D) Assertion (A) is false but reason (R) is true.	1
	SECTION B	
	Section B consists of 5 questions of 2 marks each.	
S.No		Mar ks
21	$612 = 2^2 \times 3^2 \times 17$ $1314 = 2^1 \times 3^2 \times 73^1$ $\text{HCF} = 2 \times 3 \times 3 = 18$ Or $3 \times 12 \times 101 + 4 = 4(3 \times 3 \times 101 + 1)$ $= 4(909 + 1) = 4 \times 910$ $= 2 \times 2 \times 2 \times 5 \times 7 \times 13$ is a product of prime, hence a composite number.	$\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ 1
22	Coordinates given A(5,1), B(1,4) and C(8,5) $AB = \sqrt{(1-5)^2 + (4-1)^2} = \sqrt{16+9} = \sqrt{25} = 5$ $BC = \sqrt{(8-1)^2 + (5-4)^2} = \sqrt{49+1} = \sqrt{50} = 5\sqrt{2}$ $CA = \sqrt{(5-8)^2 + (1-5)^2} = \sqrt{9+16} = \sqrt{25} = 5$ Since $AB = CA$, its an isosceles triangle Also $AB^2 + CA^2 = 5^2 + 5^2 = 50$ and $BC^2 = (\sqrt{50})^2 = 50$	1 $\frac{1}{2}$

	$AB^2 + CA^2 = BC^2$ Hence its an isosceles right-angled triangle.	1/2
23	Let the point (x,0) on x-axis divides the join of point (1,-3) and (4,5) in the ratio k:1 So, $x = \frac{4k+1}{k+1}$ and $0 = \frac{5k-3}{k+1} \Rightarrow k = \frac{3}{5}$ Hence $x = \frac{4 \times \frac{3}{5} + 1}{\frac{3}{5} + 1} = \frac{17}{8}$ Point is $(\frac{17}{8}, 0)$	1/2 1 1/2
24	$\sin\theta + \cos\theta = \sqrt{2}$ $\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 2$ (squaring both sides) $1 + 2\sin\theta\cos\theta = 2$ $\sin\theta\cos\theta = \frac{1}{2}$ Now $\tan\theta + \cot\theta = \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta} = 2$	1/2 1/2 1
25	(A) $P = \frac{260}{600} = \frac{13}{30}$ (B) $P = \frac{290}{600} = \frac{29}{60}$ Or Possible outcomes:- HHH, HHT, HTH, THH, HTT, THT, TTH, TTT (A) $P(\text{two heads}) = \frac{3}{8}$ (B) $P(\text{at most two tails}) = \frac{7}{8}$	1+1 1/2 1/2 1
	SECTION C	
	Section C consists of 6 questions of 3 marks each.	
S.No		Marks
26	(i) length of arc = $(22/7) \times 10 \times (60^\circ / 180^\circ) = (220 / 21)$ cm (ii) area of minor segment = area of sector – area of triangle = $(22/7) \times 10 \times 10 \times (60^\circ/360^\circ) - (\sqrt{3}/4) \times 10 \times 10$ (triangle will be equilateral) = $(25/21) (44 - 21\sqrt{3}) \text{ cm}^2$ Or (i) Area cleaned by each wiper = $\frac{\pi \times 3^2 \times 60^\circ}{360^\circ} = \frac{3}{2} \pi \text{ cm}^2$ (ii) Area of glass = $(\frac{5}{2})^2 \pi = \frac{25}{4} \pi \text{ cm}^2$ Percentage of area cleaned in one swipe = $(\frac{3}{2} \pi / \frac{25}{4} \pi) \times 100\% = 24\%$	1 1 1/2 1/2 1 1 1
27	Let suppose that $\sqrt{2} + \sqrt{3}$ is rational. Let $\sqrt{2} + \sqrt{3} = a$, where a is rational. Therefore, $\sqrt{2} = a - \sqrt{3}$ Squaring on both sides, we get $2 = a^2 + 3 - 2a\sqrt{3}$	1/2 1/2 1/2 1/2

	Therefore, $\sqrt{3} = \frac{a^2+1}{2a}$ which is a contradiction as the right hand side is a rational number while $\sqrt{3}$ is irrational. Hence, $\sqrt{2} + \sqrt{3}$ is irrational.	$\frac{1}{2}$ $\frac{1}{2}$
28	Let the numbers be x and $9-x$ ATQ $\frac{1}{x} + \frac{1}{9-x} = \frac{1}{2}$ $\frac{9-x+x}{(9-x)x} = \frac{1}{2}$ $9x2 = 9x - x^2$ $x^2 - 9x + 18 = 0$ $x^2 - 6x - 3x + 18 = 0$ $x(x-6) - 3(x-6) = 0$ $(x-6)(x-3) = 0$ $x = 6, \quad x = 3$ Therefore numbers are 3 and 6	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
29	 In $\triangle PAB$ and $\triangle RCB$ $\angle PAB = \angle RCB$ ($PA \parallel RB$) $\angle PAB = \angle RCB$ (common) Hence, $\triangle PAB \sim \triangle RCB$ (AA) Therefore $\frac{CB}{CA} = \frac{QB}{PA}$ $\frac{CB}{CA} = \frac{z}{x}$ -----(1) Similarly, $\triangle QAB \sim \triangle RAC$ (AA) $\frac{AB}{CA} = \frac{QB}{RC}$ $\frac{AB}{CA} = \frac{z}{y}$ -----(2) Adding (1) and (2) $\frac{AB}{CA} + \frac{CB}{CA} = \frac{z}{y} + \frac{z}{x}$ $\frac{AB+CB}{AC} = \frac{z}{y} + \frac{z}{x}$ $\frac{AC}{AC} = \frac{z}{y} + \frac{z}{x}$	1 $\frac{1}{2}$ $\frac{1}{2}$

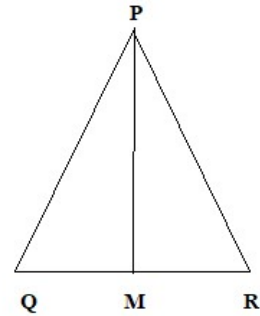
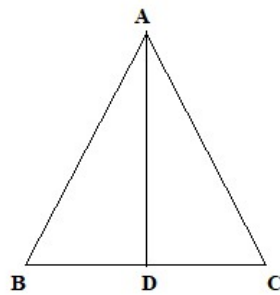
$$1 = \frac{z}{y} + \frac{z}{x}$$

$$\frac{1}{z} = \frac{1}{y} + \frac{1}{x} \quad \text{Proved}$$

Proved

 $\frac{1}{2}$

Or

 $\frac{1}{2}$

In $\triangle ABC$ and $\triangle PQR$

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$$

 $\frac{1}{2}$

$$\frac{AB}{PQ} = \frac{2BD}{2QM} = \frac{AD}{PM}$$

$$\frac{AB}{PQ} = \frac{BD}{QM} = \frac{AD}{PM}$$

So, $\triangle ABD \sim \triangle PQM$ (SSS)

$$\Rightarrow \angle B = \angle Q$$

Now in $\triangle ABC$ and $\triangle PQR$

$$\frac{AB}{PQ} = \frac{BC}{QR}$$

$$\angle ABC = \angle PQR$$

$$\triangle ABC \sim \triangle PQR(\text{SAS})$$

1

1

30

For $2x^2 - 3x + 1$

Then $\alpha + \beta = 3/2$

And $\alpha\beta = 1/2$

Now a quadratic polynomial whose zeros are 3α and 3β

$$k(x^2 - (\text{sum of zeros})x + \text{product of zeros})$$

$$= k(x^2 - (3\alpha + 3\beta)x + 3\alpha \times 3\beta)$$

$$= k(x^2 - 3(\alpha + \beta)x + 9\alpha\beta)$$

$$= k(x^2 - 3 \times \frac{3}{2}x + 9 \times \frac{1}{2})$$

$$= k(x^2 - \frac{9}{2}x + \frac{9}{2})$$

$$= 2x^2 - 9x + 9 \text{ (k taken 2)}$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

31

$$\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2\sin^2 A - 1}$$

LHS

$$\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{(\sin A + \cos A)^2 + (\sin A - \cos A)^2}{\sin^2 A - \cos^2 A}$$

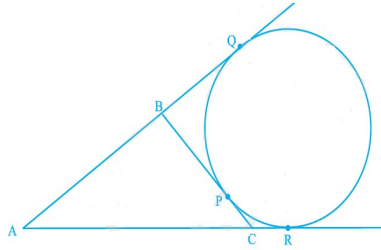
$$\frac{\sin^2 A + \cos^2 A + 2\sin A \cos A + \sin^2 A + \cos^2 A - 2\sin A \cos A}{\sin^2 A - (1 - \sin^2 A)}$$

$$\frac{1+1}{2\sin^2 A-1} = \frac{2}{2\sin^2 A-1} = \text{RHS}$$

1

1

1

	Section D consists of 4 questions of 5 marks each.	
S.No		Marks
32	Given , To prove, figure Correct proof  Lengths of tangents from an external point to a circle are equal. $BQ = BP$ $CP = CR$, and $AQ = AR$ Now, $2AQ = AQ + AR$ $= (AB + BQ) + (AC + CR)$ $= AB + BP + AC + CP$ $= (BP + CP) + AC + AB$ $= BC + CA + AB$ i.e., $AQ = \frac{1}{2}(BC + CA + AB)$	1 1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
33	Correct table of values for x and y Correct graph plotting The vertices of this triangle are B (–2, 0), Q (1, 0) and R (2, 4). Area of triangle = $\frac{1}{2} \times 3 \times 4 = 6$ sq units Or Let the cost price of the table be Rsx and the cost price of the chair be Rsy. The selling price of the table, when it is sold at a profit of 10% $= x + \frac{10x}{100} = \frac{110}{100}$ The selling price of the chair when it is sold at a profit of 25% $= y + \frac{25y}{100} = \frac{125y}{100}$ ATQ $\frac{110x}{100} + \frac{125y}{100} = 1050$ $= 110x + 125y = 105000$ -----(1) When the table is sold at a profit of 25%, its selling price $= x + \frac{25}{100} = \frac{125x}{100}$ When the chair is sold at a profit of 10%, its selling price $= y + \frac{10}{100} = \frac{110}{100}$ So, $\frac{125x}{100} + \frac{110y}{100} = 1065$	2 2 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

$$= 125x + 110y = 106500 \text{ -----}(2)$$

On adding these equations,

$$235x + 235y = 211500$$

$$\text{i.e., } x+y=900 \quad \text{-----}(3)$$

On subtracting these equations,

$$15x - 15y = 1500$$

$$\text{i.ex} - y = 100 \text{ -----(4)}$$

Solving Equations (3) and (4), we get

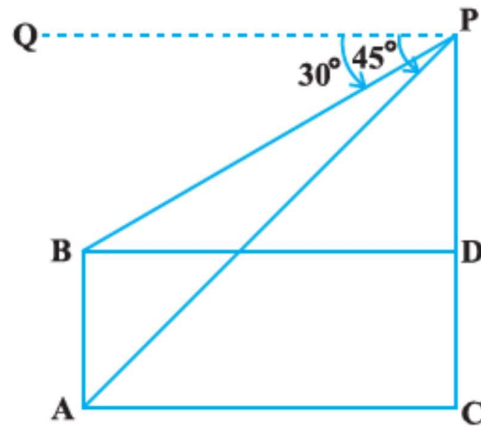
$$x = 500, y = 400$$

So, the cost price of the table is Rs 500 and the cost price of the chair is Rs 400.

 $\frac{1}{2}$ $\frac{1}{2}$

1

34



1(fig)

Let PC denotes the multi-storied building and AB denotes the 8 m tall building.

$$\angle PBD = \angle QPB = 30^\circ$$

$$\angle PAC = \angle QPA = 60^\circ$$

In right $\triangle PBD$,

$$\tan 30^\circ = \frac{PD}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{PD}{BD}$$

$$BD = \sqrt{3}PD \quad \text{-----}(1)$$

In right Δ PAC,

$$\tan 45^\circ = \frac{PC}{AC}$$

$$1 = \frac{PC}{AC}$$

$$PC = AC \quad \text{-----} (2)$$

$$PC = BD \quad (BD = AC)$$

$$PC = \sqrt{3} PD \text{ (from (1))}$$

$$PD + DC = \sqrt{3} PD$$

$$PD + 8 = \sqrt{3} PD \quad (\text{BA} = \text{DC})$$

$$PD = \frac{8}{\sqrt{3} - 1}$$

$$PD = \frac{8(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)} = 4(\sqrt{3} + 1)$$

$$\text{Height of multi- storied building} = \text{PD} + \text{DC} = 4(\sqrt{3} + 1) + 8 = 4(\sqrt{3} + 3)m$$

Also distance between two buildings = $AC = BD = PC = 4(\sqrt{3} + 3)m$

1

1

1

 $\frac{1}{2}$ $\frac{1}{2}$

35

CLASS INTERVAL	FREQUENCY	CUMULATIVE FREQUENCY
Below 140	4	4
140 - 145	7	11
145 – 150	18	29
150 – 155	11	40
155 - 160	6	46
160 - 165	5	51

$N/2 = 51/2 = 25.5$
Median class = 145 – 150
$$\text{Median} = l + \left(\frac{\frac{N}{2} - CF}{f} \right) \times h$$
$$= 145 + \frac{(25.5 - 11)}{18} \times 5$$
$$= 149.03 \text{ cm}$$

or

Age (in years)	Number of patients f_i	Class mark x_i	$di = xi - 30$	$fidi$
5 – 15	6	10	-20	-120
15 – 25	11	20	-10	-110
25 – 35	21	30	0	0
35 – 45	23	40	10	230
45 – 55	14	50	20	280
55 – 65	5	60	30	150
TOTAL	80			430

Mean- Let assumed mean be 30
$$\text{Mean} = a + \frac{\sum fd}{\sum f}$$
$$30 + \frac{430}{80}$$
$$30 + 5.375$$
$$35.375 \text{ years}$$

Mode
maximum class frequency is 23 belonging to class interval 35 - 45.
Modal class = 35 - 45
Lower limit (l) of modal class = 35
Frequency (f_1) of modal class = 23
Class size (h) = 10
Frequency (f_0) of class preceding the modal class = 21
Frequency (f_2) of class succeeding the modal class = 14
$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$
$$= 35 + \left(\frac{23 - 21}{2 \times 23 - 21 - 14} \right) \times 10$$
$$= 35 + 1.818$$

2

$\frac{1}{2}$

$1 \frac{1}{2}$

1

$1 \frac{1}{2}$

1

1

	= 36.82 years Hence median using empirical formula $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$ $\text{Median} = \frac{(36.82 + 2 \times 35.36)}{3} = 35.85$	1/2
	SECTION E	
	Case study based questions are compulsory.	
36	(i) Slant height = $\sqrt{8^2 + 15^2} = \sqrt{289} = 17m$ (ii) Area of canvas = CSA of cylinder + CSA of cone $= 2 \times \frac{22}{7} \times 15 \times 9 + \frac{22}{7} \times 15 \times 17 = 1650 m^2$ (iii) The cost of canvas = $(1650 + 30) \times 200$ $= \text{Rs } 336000$ Or Volume of air = $\pi r^2 h + \frac{1}{3} \pi r^2 H$ $= \frac{22}{7} \times 15^2 \times 9 + \frac{1}{3} \times \frac{22}{7} \times 15^2 \times 8$ $= 8250 m^3$	1 1 1 1 1 1
37	(i) Yes $\triangle AEB \sim \triangle CED$ as $\angle E = \angle E$ (common) $\angle EDC = \angle EBA$ (corresponding angles) $\triangle AEB \sim \triangle CED$ (AA) (ii) $\frac{H}{h} = \frac{L}{l}$ $\frac{3}{1} = \frac{6}{l}$ $l = 2m$ (iii) Using pythagoras theorem in $\triangle ABE$ $8^2 + 6^2 = BE^2$ $100 = BE^2$ $BE = 10 m$ Or $\frac{H}{h} = \frac{L}{l}$ $\frac{4}{1} = \frac{6}{l}$ $l = 1.5 m$	1 1 2
38	(i) $a + 5d = 16000$ $a + 8d = 22600$ Solving both equations $d = 2200$, $a = 5000$ So production in first year was 5000. (ii) Total production during first 3 years = $\frac{3}{2} (2 \times 5000 + 2 \times 2200) = 21600$ (iii) Let in nth year production was 29200 $29200 = 5000 + (n-1)2200$ $n = 12$ or Difference in the production in 7th and 4th year $= a + 6d - (a + 3d) = 3d$ $= 3 \times 2200 = 6600$	1 1 1 1 1 1